

DEEKSHA REDDY PATLOLLA

Data Engineer | Denver, CO | (440) 755-0232
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SUMMARY

Data Engineer who designs and ships end-to-end ETL/ELT pipelines, dimensional data warehouses, and ML-integrated analytics using Python, SQL, Apache Spark, and Airflow. Builds automated, observable, cloud-native data workflows on Azure and translates stakeholder requirements into reliable, production-ready data models. Master's in Computer Science with strengths in distributed data processing, query optimization, and data-quality engineering.

EXPERIENCE

Data Engineer

Mar 2025 – May 2026

University of Colorado, Denver | Denver, CO

- **Designed and built** end-to-end ETL pipelines in Python, SQL, and Apache Spark, ingesting and transforming data from multiple institutional source systems to power research analytics at scale.
- **Automated** data-ingestion workflows with Apache Airflow, cutting manual processing time by 50% and ensuring consistent data freshness for downstream dashboards and reports.
- **Architected** star- and snowflake-schema data warehouse structures, improving query performance and simplifying reporting access for analysts.
- **Optimized** Spark jobs and complex SQL queries by profiling processing bottlenecks, reducing job runtimes and increasing throughput across the analytics platform.
- **Engineered** automated data-validation checks and schema-drift alerts inside pipelines, catching anomalies early and safeguarding downstream data integrity.
- **Integrated** Scikit-learn classification models on normalized relational schemas to predict student performance, enabling proactive, data-driven academic advising.
- **Partnered with** advisors and administrators to translate reporting requirements into technical data models and pipeline specs, and authored ETL documentation and data dictionaries to speed onboarding.

Environment: Python, SQL, Apache Spark, Apache Airflow, Scikit-learn, PostgreSQL, Azure SQL, Azure Monitor, Application Insights, Data Warehousing

Software Engineer Intern

Aug 2023 – May 2024

Infor Pvt Ltd | Hyderabad, India

- **Developed and maintained** Java/Spring Boot microservices and RESTful endpoints for enterprise ERP modules, contributing to improved service reliability.
- **Wrote and optimized** SQL queries, indexes, and stored procedures in Azure SQL and PostgreSQL, reducing query response times across high-traffic endpoints.
- **Built and maintained** CI/CD pipelines in Azure DevOps and GitHub Actions, automating build, test, and deployment; containerized services with Docker for Azure App Service and AKS.
- **Implemented** authentication and authorization with Azure Entra ID and OAuth2/OIDC in Spring Boot, using Azure Key Vault for secrets management.
- **Authored** unit and integration tests with JUnit and Mockito, raising code coverage on core modules and improving release confidence within Agile sprints.

Environment: Java, Spring Boot, Spring MVC/Data/Security, REST APIs, Azure (App Service, AKS, Entra ID, Key Vault), Azure DevOps, GitHub Actions, Docker, JUnit, Mockito, Azure SQL, PostgreSQL, Git, Maven

TECHNICAL SKILLS

Programming	Python, SQL, Java (8/11/17), C/C++, JavaScript, Bash
Data & Big Data	Apache Spark, Apache Airflow, Kafka, ETL/ELT, Data Warehousing, Star/Snowflake Schemas, Redis
Databases	PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Cassandra
Cloud & DevOps	Azure (SQL, App Service, AKS, Service Bus, Event Hubs, Monitor, App Insights, Entra ID, Key Vault), AWS, Docker, Kubernetes, Azure DevOps, GitHub Actions, CI/CD, Linux
Backend	Spring Boot, Spring (MVC/Data/Security), Hibernate, JPA, FastAPI, REST APIs, GraphQL
AI / ML	PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, LangChain, RAG, pgvector

PROJECTS

<u>CivicLens – Policy Simulation Platform (4th Place, LynxHack)</u>	Apr 2026
• Built a full-stack agentic AI platform that ingests live data from the US Census and FRED APIs and runs three parallel AI agents (Economist, Urban Planner, Equity Analyst) to simulate policy impact. • Engineered a RAG pipeline (ChromaDB, sentence-transformers) with LangGraph orchestration and a 3D Mapbox UI (Next.js/TypeScript); reduced response latency from ~45s to ~15s via parallel agent execution.	
<u>Heterogeneity-Aware Multimodal Learning for Early Disease Detection</u>	Dec 2025
• Built an end-to-end PyTorch training pipeline fusing imaging, genomic, and clinical data with a transformer fusion model, improving ROC-AUC by 11% over early-fusion baselines. • Added gated attention, cross-validation, and SHAP-based interpretability for modality-level contribution analysis.	
<u>Resume AI Intelligence Platform (RAG + Multi-Agent LLM)</u>	Nov 2025
• Designed a multi-agent LLM system (LangChain + GPT-4) with a RAG pipeline on pgvector/Supabase for semantic resume parsing and job matching. • Built a production-ready FastAPI backend with memory-aware context handling, reducing token usage by 28%.	
<u>Autonomous Cybersecurity Defender</u>	Sep 2025
• Built a real-time intrusion-detection system combining rule-based filters with GPT-4o classification on the CICIDS2017 dataset, reaching 94% detection accuracy; served via a FastAPI backend at ~2.5s latency.	

EDUCATION

M.S., Computer Science - GPA 3.8 <i>University of Colorado, Denver Denver, CO</i>	Aug 2024 – May 2026
B.Tech, Computer Science - GPA 3.6 <i>Malla Reddy College of Engineering and Technology Hyderabad, India</i>	Aug 2020 – May 2024

PUBLICATIONS

<u>Unmasking Digital Deception: Detecting Deepfakes through Deep Learning</u> — <i>IRJMETs, 2024</i>
• Developed a hybrid CNN-RNN architecture (ResNeXt + LSTM) capturing spatial and temporal inconsistencies in video, with a multi-dataset training pipeline (DFDC, FaceForensics++, Celeb-DF) for improved generalization.